# MoonEP and Perfectly Balanced Expert Parallelism

Deep Dive into Kimi K3 — Chapter 37

Yin Yang

Foundation Books Project

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Goal: type the distributed route contract, prove the bounded-copy balance theorem, plan online, keep temporary values fresh and views leased, return every gradient to its owner, and keep each measurement in scope.

# Roadmap

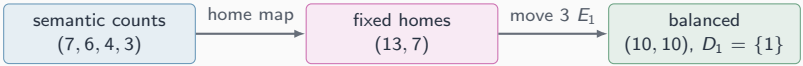
Balance columns, preserve rows

Fresh values, layouts, and transport

Owners, envelopes, and evidence

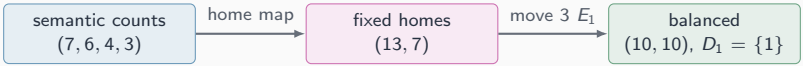
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- An exact oracle minimizes worst-rank remote types; K3 reports a near-optimal online GPU planner. [REPORTED: K3 Sec. 5.2.1]

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- Symmetric $[E+B, H, H']$ views map home rows and add $B$ local slots; zero-copy removes a materialization, not traffic or synchronization.
- Exact balance fixes $SK$ real routes per rank — not per-expert groups, padding, or elapsed time.

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Balance is counts, correctness is identity, speed is evidence. Next: Chapter 38's Distributed Execution for K3-Class Models.